“Familiarization of Seebeck Effect and Characteristic Study of Peltier Module”

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“FAMILIARIZATION OF SEEBECK EFFECT AND CHARACTERISTIC STUDY OF PELTIER MODULE”

INDEX

Chapter 5 : CONCLUSION

- Advantages of output voltages
- Future aspects of Peltier module
- Reference

CHAPTER 1

INTRODUCTION

Abstract

Energy is one of the basic needs of our lives. Thermal energy is the first form of energy familiarized by humans with the invention of fire. From then, different types of energy conversions were involved for their existence. While cooking animal flesh, the thermal energy was converted into chemical energy, while pulling carts, mechanical energy was converted into kinetic energy and so on.Through different human civilizations upto the modern era all inventions and technologies destined to facilitate the human life.Electrical energy became the most demanded form of energy with industrialization. On large scale electrical energy is produced from different sources like running water, wind, tide, thermal and nuclear energy. Due to the scarcity of non-renewable energy sources humans have started to find alternative energy sources. In literature it can be seen that various methodologies and theories have been introduced in this regard. Faraday's law of induction, Lenz's law, Law of conservation of energy and Joule's law are some of them. Electrical energy can be tapped from the renewable energy sources like solar energy, thermal energy, nuclear energy, hydro power energy, wind energy etc. All the above-mentioned energy sources require heavy cost, maintenance and immense labor which paved the way to various research in this field to find an alternative source for the electrical energy which can produce electric current or voltage by simple mechanism and at low cost. Thermoelectricity the direct conversion of thermal energy to electrical energy or vice versa was introduced in the 18th century. But its practical implementation was done with the invention of semiconductors and related effects. Thermoelectricity actually is composed of three major inventions or effects - Seebeck effect, Peltier effect and Thomson effect.

In this project work we introduce a characteristic study of a Peltier module and familiarization of Seebeck effect is done. The parametric study of Peltier module may be useful for researchers in this field. By combining a large number of Peltier modules a new source of electrical energy can be produced.

Thermoelectricity

Thermoelectricity offers an efefctive path to recover and convert waste heat into readily available electric energy, and has been studied for more than two centuries. From the controversy between Galvani and Volta on the Animal Electricity, dating back to the end of the 18th century and anticipating Seebeck's observations, the understanding of the physical mechanisms evolved along with the development of the technology. In the 19th century Orsted clarified some of the earliest observations of the thermoelectric phenomenon and proposed the first thermoelectric pile, while it was only after the studies on thermodynamics by Thomson, and Rayleigh's suggestion to exploit the Seebeck effect for power generation, that a diverse set of thermoelectric generators was developed. From such pioneering endeavors, technology evolved from massive, and sometimes unreliable, thermopiles to very reliable devices for sophisticated niche applications in the 20th century, when Radioisotope Thermoelectric Generators for space missions and nuclear batteries for cardiac pacemakers were introduced. While some of the materials adopted to realize the first thermoelectric generators are still investigated nowadays, novel concepts and improved understanding of materials growth, processing, and characterization developed during the last 30 years have provided new avenues for the enhancement of the thermoelectric conversion efficiency, for example through nano structuration, and favoured the development of new classes of thermoelectric materials. With increasing demand for sustainable energy conversion technologies, the latter aspect has become crucial for developing thermoelectrics based on abundant and non-toxic materials, which can be processed at economically viable scales, tailored for different ranges of temperature. This includes high temperature applications where a substantial amount of waste energy can be retrieved, as well as room temperature applications where small and local temperature differences offer the possibility of energy scavenging, as in micro harvesters meant for distributed electronics such as sensor networks. While large scale applications have yet to make it to the market, the richness of available and emerging thermoelectric technologies presents a scenario where thermoelectrics is poised to contribute to a future of sustainable future energy harvesting and management.

Thermoelectric Couple

A **thermocouple**, also known as a "thermoelectrical thermometer", is an electrical device consisting of two dissimilar electrical conductors forming an electrical junction. A thermocouple produces a temperature-dependent voltage as a result of the Seebeck effect, and this voltage can be interpreted to measure temperature. Thermocouples are widely used as temperature sensors.

Thermocouples are widely used in science and industry. Applications include temperature measurement for kilns, gas turbine exhaust, diesel engines, and other industrial processes. Thermocouples are also used in homes, offices and businesses as the temperature sensors in thermostats, and also as flame sensors in safety devices for gas-powered appliances. It works on the principle of thermodynamics and this effect is known as thermoelectric effect. The thermoelectric effect is the direct conversion of temperature differences to electric voltage and vice versa via a thermocouple. A thermoelectric device creates a voltage when there is a different temperature on each side. Conversely, when a voltage is applied to it, heat is transferred from one side to the other, creating a temperature difference. At the atomic scale, an applied temperature gradient causes charge carriers in the material to diffuse from the hot side to the cold side. This effect can be used to generate electricity, measure temperature or

change the temperature of objects. Because the direction of heating and cooling is affected by the applied voltage, thermoelectric devices can be used as temperature controllers

Thermoelectric couples are based on three laws:

* Seebeck effect

*Peltier effect

*Thomson effect

Seebeck Effect

Seebeck effect was introduced by Thomas Johann Seebeck in 1821. The Seebeck effect is the electromotive force (emf) that develops across two points of an electrically conducting material when there is a temperature difference between them. The emf is called the Seebeck emf (or thermos /thermal /thermoelectric emf). The ratio between the emf and temperature difference is the Seebeck coefficient.

A thermocouple measures the difference in potential across a hot and cold end for two dissimilar conductors or semi-conductors. This potential difference is proportional to the temperature difference between the hot and cold ends.

The generated voltage (v) is the Seebeck voltage and is related to the difference in (dt) between the heated junction and the open junction by a proportional factor called the Seebeck coefficient. (v=Seebeck coefficient x dt) Seebeck observed what he called "thermomagnetic effect" wherein a magnetic compass needle would be deflected by a closed loop formed by two different metals joined in two places, with an applied temperature difference between the joints. The Seebeck effect is a reversible process , if the hot and cold junctions are interchanged then the direction of the current will also change, therefore we can say that the thermoelectric process is a reversible process. The SI unit of Seebeck effect is volts/kelvin.

Seebeck effect is caused due to the higher energy of the electron on the warmer side while the electrons on lower side have less energy. This results in a concentration of the electrons in thecooler side which results in increased negative charge on the cooler side. Thermocouples involve two wires, each of a different material, that are electrically joined in a region of unknown temperature. The loose ends are measured in an open-circuit state and as a result we will get the output voltage called Seebeck voltage. Thermoelectric generators are like a thermocouple.

Peltier Effect

The Peltier effect is the reverse phenomenon of Seebeck effect and it was found by French physicist Jean Charles Athanase Peltier, who discovered it in 1834. When an electric current is passed through a circuit of a thermocouple, heat is generated at one junction and absorbed at the other junction. This is known as the **Peltier effect**: the presence of heating or cooling at an electrified junction of two different conductors. When the electrical current flowing through the junction connecting two materials will emit or absorb heat per unit time at the junction to balance the difference in the chemical potential of the two materials. When the current flows the heat is deposited in one junction and cooling occurs in another junction. The main application of Peltier effect is cooling. Peltier module is a device which works on the principle of Peltier effect. When the electricity is passed through this module the electrons move in one element and positive holes move in other element, this allows one side of the substrate to absorb heat and the other to radiate heat. The Peltier module is a reversible device which means that depending on the current direction the hot and the cold sides of the module can be switched hence Peltier module is a reversible device and Peltier effect is a reversible process. The

Peltier effect works on the DC voltage it is due to maintain the direction of heat flow because if AC voltage is used the direction of heat changes with the direction of current flow.

The application of Peltier effect is The Peltier effect can be used to create a heat pump. Notably, the Peltier thermoelectric cooler is a refrigerator that is compact and has no circulating fluid or moving parts. Such refrigerators are useful in applications where their advantages outweigh the disadvantage of their very low efficiency.

Other heat pump applications such as dehumidifiers may also use Peltier heat pumps.

Thermoelectric coolers are trivially reversible, in that they can be used as heaters by simply reversing the current. Unlike ordinary resistive electrical heating (Joule heating) that varies with the square of current, the thermoelectric heating effect is linear in current (at least for small currents) but requires a cold sink to replenish with heat energy. This rapid reversing heating and cooling effect is used by many modern thermal cyclers, laboratory devices used to amplify DNA by the polymerase chain reaction (PCR). PCR requires the cyclic heating and cooling of samples to specified temperatures. The inclusion of many thermocouples in a small space enables many samples to be amplified in parallel.

Thomson Effect

In different materials, the Seebeck coefficient is not constant in temperature, and so a spatial gradient in temperature can result in a gradient in the Seebeck coefficient. If a current is driven through this gradient, then a continuous version of the Peltier effect will occur. This **Thomson effect** was predicted and later observed in 1851 by William Thomson. It describes the heating or cooling of a current-carrying conductor with a temperature gradient. If a current density is passed through a homogeneous conductor, the Thomson effect predicts a heat production rate per unit volume. The Thomson effect is a manifestation of the direction of flow of electrical carriers with respect to a temperature gradient within a conductor. These absorb energy (heat) flowing in a direction opposite to a thermal gradient, increasing their potential energy, and, when flowing in the same direction as a thermal gradient, they liberate heat, decreasing their potential energy.

The above mentioned three laws are the basis of thermoelectric couples.

The working or mode of operation of thermocouple is as follows, In 1821, the German physicist Thomas Johann Seebeck discovered that a magnetic needle held near a circuit made up of two dissimilar metals got deflected when one of the dissimilar metal junctions was heated. At the time, Seebeck referred to this consequence as thermo-magnetism. The magnetic field he observed was later shown to be due to thermo-electric current. In practical use, the voltage generated at a single junction of two different types of wire is what is of interest as this can be used to measure temperature at very high and low temperatures. The magnitude of the voltage depends on the types of wire being used. Generally, the voltage is in the microvolt range and care must be taken to obtain a usable measurement. Although very little current flows, power can be generated by a single thermocouple junction. Power generation using multiple thermocouples, as in a thermopile, is common.

Thermopile is known as thermopile is an electronic device that converts thermal energy into electrical energy. It is composed of several thermocouples connected usually in series or, less commonly, in parallel. Such a device works on the principle of the thermoelectric effect, i.e., generating a voltage when its dissimilar metals are exposed to a temperature difference. Thermocouples can be connected in series as thermocouple pairs with a junction located on either side of a thermal resistance layer. The output from the thermocouple pair will be a voltage that is directly proportional to the temperature difference across the thermal resistance layer and also to the heat flux through the thermal resistance layer. Adding more thermocouple pairs in series increases the magnitude of the voltage output. Thermopiles can be constructed with a single thermocouple pair, composed of two thermocouple junctions, or multiple thermocouple pairs.

Scope Of Thermoelectric Couples (Heater & Cooler)

Peltier module is a type of thermoelectric couple which is used to emit heat and the other side is used to absorb heat, so always one side will be always heated and opposite side be cooler. So Peltier module has so much scope in real world;

Using Peltier module we can construct a refrigerator in which the substances can be cooled up to satisfactory level.

Ophthalmic Lasers: Inside Ophthalmic Lasers, Peltier is used to keeping the laser diodes below a certain temperature, and cool the Crystals.

Night Vision Devices: To avoid image noise, image sensors are cooled using the Peltier.

Battery Cooling of Electric Cars: One can use Peltiers to Cool Li-Ion batteries of Electric Cars.

Cryopreservation: We can use Peltier to preserve tissues, cells, and other similar substances in cool temperatures that are cooled using Peltier modules.

One of the other major use of thermoelectric couple/ thermoelectric heater, is that it can be used roofings of houses and buildings which will help to maintain the room temperature. Because by using thermoelectric coolers or heaters one side will always absorbs the heat and the other side will always emit heat hence in colder countries , inside room, temperature can be maintained at low cost and at low maintenance.

CHAPTER 2
PELTIER MODULE

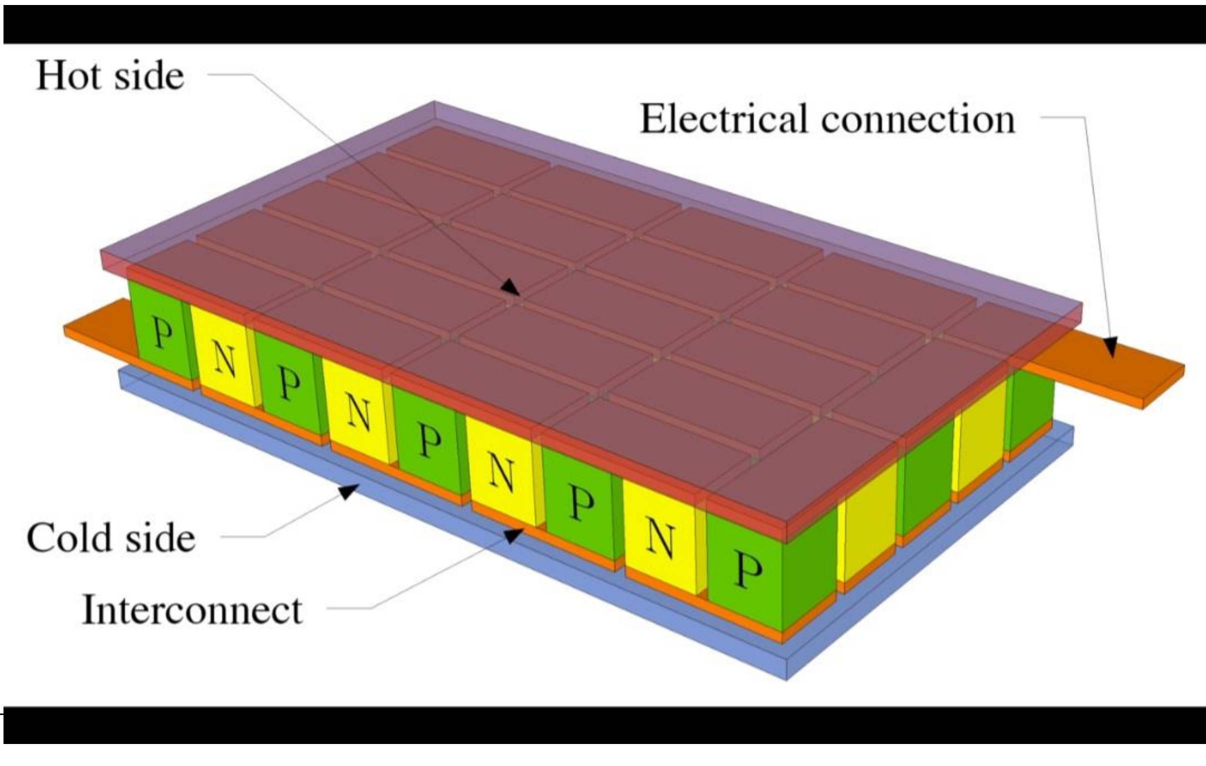
Thermoelectric Cooling

Thermoelectric cooling uses the Peltier effect to create a **heat** flux at the junction of two different types of materials. A Peltier cooler, heater, or **thermoelectric** heat pump is a **solid-state** active heat pump which transfers heat from one side of the device to the other, with consumption of electrical energy, depending on the direction of the current. Such an instrument is also called a **Peltier device**, **Peltier heat pump**, **solid state refrigerator**, or **thermoelectric cooler** (**TEC**) and occasionally a **thermoelectric battery**. It can be used either for heating or for cooling, although in practice the main application is cooling. It can also be used as a temperature controller that either heats or cools.

This technology is far less commonly applied to refrigeration than vapour compression refrigeration is. The primary advantages of a Peltier cooler compared to a vapour-compression refrigerator are its lack of moving parts or circulating liquid, very long life, invulnerability to leaks, small size, and flexible shape. Its main disadvantages are high cost for a given cooling capacity and poor power efficiency (a low coefficient of performance or COP). Many researchers and companies are trying to develop Peltier coolers that are cheap and efficient.

A Peltier cooler can also be used as a thermoelectric generator. When operated as a cooler, a voltage is applied across the device, and as a result, a difference in temperature will build up between the two sides. When operated as a generator, one side of the device is heated to a temperature greater than the other side, and as a result, a difference in voltage will build up between the two sides (the Seebeck effect). However, a well-designed Peltier cooler will be a mediocre thermoelectric generator and vice versa, due to different design and packaging requirements.

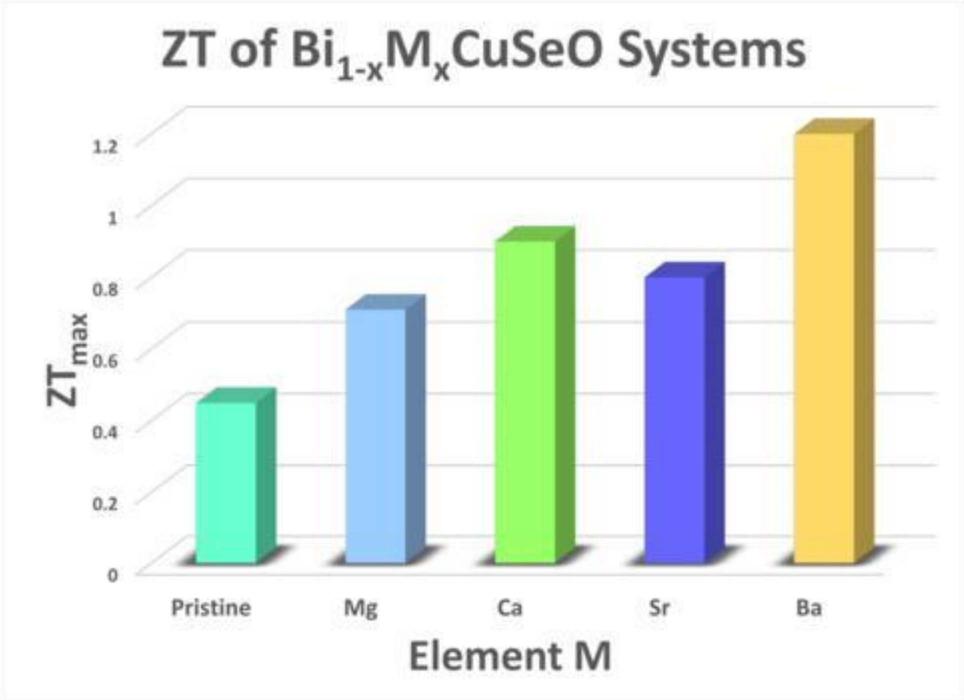
Structure



Design

Two unique semiconductors, one n-type and one p-type, are used because they need to have different electron densities. The alternating p & n-type semiconductor pillars are placed thermally in parallel to each other and electrically in series and then joined with a thermally conducting plate on each side, usually ceramic, removing the need for a separate insulator. When a voltage is applied to the free ends of the two semiconductors there is a flow of DC current across the junction of the semiconductors, causing a temperature difference. The side with the cooling plate absorbs heat which is then transported by the semiconductor to the other side of the device.

The cooling ability of the total unit is then proportional to the total cross section of all the pillars, which are often connected in series electrically to reduce the current needed to practical levels. The length of the pillars is a balance between longer pillars, which will have a greater thermal resistance between the sides and allow a lower temperature to be reached but produce more resistive heating, and shorter pillars, which will have a greater electrical efficiency but let more heat leak from the hot to cold side by thermal conduction. For large temperature differences, longer pillars are far less efficient than stacking separate, progressively larger modules; the modules get larger as each layer must remove both the heat moved by the above layer and the waste heat of the layer.



Materials

Requirements for thermoelectric materials:

- Narrow band-gap semiconductors because of room-temperature operation;
- High electrical conductivity (to reduce electrical resistance, a source of waste heat);

- Low thermal conductivity (so that heat doesn't come back from the hot side to the cool side);
- Large unit cell, complex structure;
- Highly anisotropic or highly symmetric;
- Complex compositions.

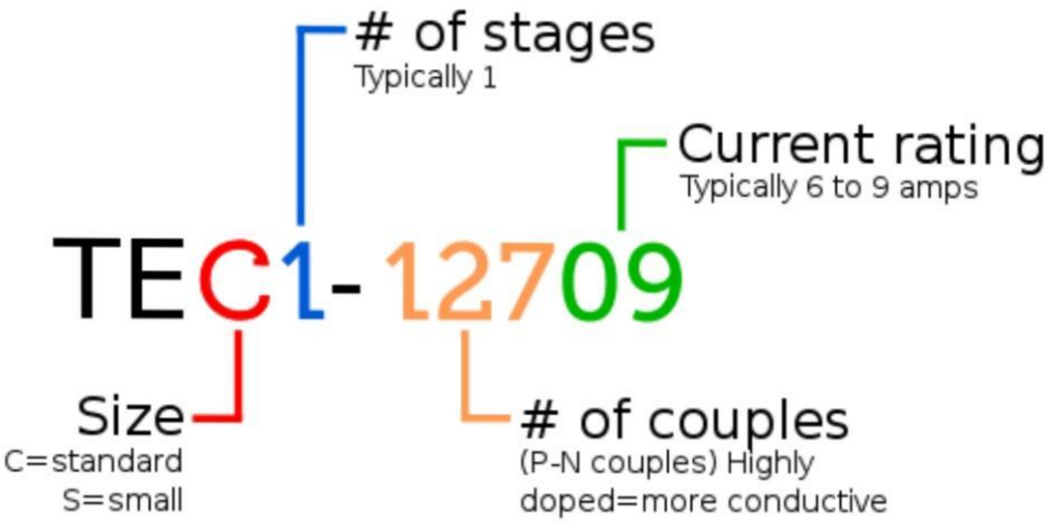
Materials suitable for high efficiency TEC systems must have a combination of low thermal conductivity and high electrical conductivity. The combined effect of different material combinations is commonly compared using a figure of merit known as ZT, a measure of the system's efficiency. The equation for ZT is given below, where alpha is the seedback coefficient , sigma is the electrical conductivity and kappa is the thermal conductivity

{T} =(alpha ^{2}\sigma {T})/kappa }

There are few materials that are suitable for TEC applications since the relationship between thermal and electrical conductivity is usually a positive correlation. Improvements in reduced thermal transport with increased electrical conductivity are an active area of material science research. Common thermoelectric materials used as semiconductors include bismuth telluride, lead telluride, silicon–germanium, and bismuth antimonide alloys. Of these, bismuth telluride is the most commonly used. New high-performance materials for thermoelectric cooling are being actively researched.

For decades, narrow bandgap semiconductors, such as bismuth, tellurium and their compounds, have been used as materials of thermocouples.

Identification



The vast majority of thermoelectric coolers have an ID printed on the cooled side.These universal Ids clearly indicate the size, number of stages, number of couples, and current rating in amps, as seen in the adjacent diagram.

Very common Tec1-12706, square of 40 mm size and 3–4 mm high, are found for a few dollars, and sold as able to move around 60 W or generate a 60 °C temperature difference with a 6 A current. Their electrical resistance will be of 1–2 ohm magnitude.

Advantage And Disadvantage

There are many factors motivating further research on TEC including lower carbon emissions and ease of manufacturing. However, several challenges have arisen.

Benefits:

A significant benefit of TEC systems is that they have no moving parts. This lack of mechanical wear and reduced instances of failure due to fatigue and fracture from mechanical vibration and stress increases the lifespan of the system and lowers the maintenance requirements. Current technologies show the mean time between failures (MTBF) to exceed 100,000 hours at ambient temperatures.

The fact that TEC systems are current-controlled leads to another series of benefits. Because the flow of heat is directly proportional to the applied DC current, heat may be added or removed with accurate control of the direction and amount of electric current. In contrast to methods that use resistive heating or cooling methods that involve gases, TEC allows for an equal degree of control over the flow of heat (both in and out of a system under control). Because of this precise bidirectional heat flow control, temperatures of controlled systems can be precise to fractions of a degree, often reaching precision of milli Kelvin (mK) in laboratory settings.

TEC devices are also more flexible in shape than their more traditional counterparts. They can be used in environments with less space or more severe conditions than a conventional refrigerator. The ability to tailor their geometry allows for the delivery of precise cooling to very small areas. These factors make them a common choice in scientific and engineering applications with demanding requirements where cost and absolute energy efficiency are not primary concerns.

Another benefit of TEC is that it does not use refrigerants in its operation. Prior to their phaseout some early refrigerants, such as chlorofluorocarbons (CFCs), contributed significantly to ozone depletion. Many refrigerants used today also have significant environmental impact with global warming potential or carry other safety risks with them.

Disadvantages

TEC systems have a number of notable disadvantages. Foremost is their limited energy efficiency compared to conventional vapor-compression systems and the constraints on the total heat flux (heat flow) that they are able to generate per unit area.

Performance

Peltier (thermoelectric) performance is a function of ambient temperature, hot and cold side heat exchanger (heat sink) performance, thermal load, Peltier module (thermopile) geometry, and Peltier electrical parameters.

The amount of heat that can be moved is proportional to the current and time that is $Q=PIt$, where P is the Peltier coefficient, I is the current, and t is the time. The Peltier coefficient depends on temperature and the materials the cooler is made of. Magnitude of 10 watt per ampere are common, but this is offset by two phenomena:

According to Ohm's law, a Peltier module will produce waste heat itself,

Q_{waste}=RI^{2}t}, where R is the resistance.

Heat will also move from the hot side to the cool side by thermal conduction inside the module itself, an effect which grows stronger as the temperature difference grows.

The result is that the heat effectively moved drops as the temperature difference grows, and the module becomes less efficient. There comes a temperature difference when the waste heat and heat moving back overcomes the moved heat, and the module starts to heat the cool side instead of cooling it further. A single-stage thermoelectric cooler will typically produce a maximal temperature difference of 70 °C between its hot and cold sides.

Another issue with performance is a direct consequence of one of their advantages: being small. This means that:

The hot side and the cool side will be very close to each other (a few millimeters away), making it easier for the heat to go back to the cool side, and harder to insulate the hot and cool side from each other

A common 40 mm × 40 mm can generate 60 W or more—that is, 4 W/cm2 or more—requiring a powerful radiator to move the heat away

In refrigeration applications, thermoelectric junctions have about ¼ the efficiency compared to conventional means (vapor compression refrigeration): they offer around 10–15% efficiency (COP of 1.0–1.5) of the ideal Carnot cycle refrigerator, compared with 40–60% achieved by conventional compression-cycle systems (reverse Rankine systems using compression/expansion).[14] Due to this lower efficiency, thermoelectric cooling is generally only used in environments where the solid-state nature (no moving parts), low maintenance, compact size, and orientation insensitivity outweighs pure efficiency.

While lower than conventional means, efficiency can be good enough, provided:

Temperature difference is kept as small as possible, and,

The current is kept low, because the ratio of moved heat over waste heat (for same temperature on the hot and cool side) will be

{\frac {Q}{Q_{waste}}}={\frac {P}{RI}}}.

However, since low current also means a low amount of moved heat, for all practical purposes the coefficient of performance will be low.

Uses

Thermoelectric coolers are used for applications that require heat removal ranging from milliwatts to several thousand watts. They can be made for applications as small as a beverage cooler or as large as a submarine or railroad car. TEC elements have limited life time. Their health strength can be measured by the change of their AC resistance (ACR). As a cooler element wears out, the ACR will increase.

CHAPTER: 3

EXPERIMENTAL SET-UP AND METHODOLOGY

To verify the working of Peltier module and Seebeck effect we divided and decided to conduct the experiment with various aims.

The different aims were:

1. Verification of Seebeck effect

The remaining experiments were to verify the working of the Peltier module , the experiments were as follow:

1. Flame displacement – voltage study
2. Surface area – voltage study
3. Sunlight – voltage study

These were the different aims which we used to study the Seebeck effect and the Peltier effect. We already mentioned that peltier module converts heat into voltage, through which we can make the devices run.

The experimental setup to verify our aim is as follows , first we will verify the Seebeck effect and after that we will use the Peltier module to verify our rest of aims.

SEEBECK EFFECT MEASUREMENTS.

AIM

Our aim is to verify the Seebeck effect using the experimental setup.

APPARATUS REQUIRED

Copper, Aluminium, Iron , Multimeter, Flame (to supply heat)

THEORY

The Seebeck effect is the electromotive force (emf) that develops across two points of an electrically conducting material when there is a temperature difference between them.

The EMF thus generated is known as Seebeck EMF or Seebeck voltage.

PROCEDURE

To start with our experiment to verify the Seebeck effect we need to create a junction between the two different conducting materials. We will first start with copper and aluminium (strip). The both copper and aluminium is connected by twisting the ends and a junction is formed between them, now this junction is heated and the other two ends are placed in normal temperature the generated voltage output is measured using the multimeter. Following this experiment the materials used were changed and the experiment is repeated for the following materials also.

The corresponding voltage time graph is plotted.

OBSERVATIONS

For copper and aluminium(strip)

VOLTAGE (mV)	TIME(s)
0.9	1
1	2
1.2	3
1.5	4
1.6	5

Similarly the experiment is repeated for the copper and aluminium (rod)

OBSERVATIONS

VOLTAGE (mV)	TIME(s)
0.5	1
0.6	2
0.7	3
0.8	4
1	5

The experiment is repeated for iron and copper

OBSERVATIONS

VOLTAGE (mV)	TIME(s)
15	1
18	2
20	3
26	4
30	5

Fron the graph (voltage-time graph) we can see the differences between different materials and the corresponding voltage produced by them.

Now we will start the experiments to verify our peltier module.

Our aims were:

1.Flame displacement-voltage study

2.Surface area – voltage study

3.Sunlight – voltage study

FLAME DISPLACEMENT – VOLTAGE STUDY

AIM

Our aim is to verify the corresponding working of Peltier module with different regions in the flame.

APPARTUS REQUIRED

Flame, Peltier Module , Multimeter.

Theory

A flame has three parts, they are as follows:

1. Innermost zone

2. Middle zone
3. Outermost zone

The above three are the three different zones in the flame, hence in this three regions will have different temperatures. The innermost zone will have least heat when compared to the other two regions in the flame. Because it receives only least amount of oxygen to burn. The middle zone of the flame will burn by producing more heat when compared with the innermost zone, because it receives more oxygen when compared with the innermost zone. The outermost region will give more temperature because in this region the complete combustion of air will take place there by producing larger amount of heat. So our aim is to determine the voltages generated in this three different regions.

PROCEDURE

As discussed in the theory we will divide the flame in to three regions (imaginary lines are used). The innermost region will be near the core , and the middle region will be in the middle, and the outermost region will be in the outermost part. Consider if the total displacement of flame is 2cm, then the innermost region will be at the starting point say 0.2 cm then the middle region will be around the 1cm, and the outermost region lies around the0.8 - 2.0 cms respectively.

To start the experiment the Peltier module is bring close to the innermost regions and the corresponding voltage output has been taken from the Peltier module using a multimeter, and it is been marked. Similarly like the above mentioned way the corresponding readings in the middle zone region and the outermost region is also measured.

Finally graphs are drawn with the time and voltage measured.

OBSERVATIONS (INNERMOST REGIONS)

TIME (s)	OBSERVED VOLTAGE (mV)
1	20
2	23
3	26
4	30
5	36
6	42

OBSERVATIONS (MIDDLE REGION)

TIME (s)	OBSERVED VOLTAGE (mV)
1	50
2	55
3	60
4	63
5	65
6	67

OBSERVATION (OUTERMOST REGION)

TIME (s)	OBSERVED VOLTAGE (mV)
1	70
2	75
3	80
4	85
5	90
6	97

From these observations the corresponding graphs are drawn.

SURFACE AREA – VOLTAGE STUDY

AIM

The surface area – voltage relation between the Peltier module and the output voltage generated is measured.

APPARATUS REQUIRED

Peltier module, Flame, Cardboard (bad conductor of heat), Multimeter.

THEORY

The Peltier module produces the output voltage when heat is given.

PROCEDURE

The Peltier module gives output voltage when heat is supplied to it, now we will study how the output voltage is related to the surface area.

For that we will place a bad conductor between the Peltier module and the flame, for pratical purposes we will use cardboard which is a insulator, for the verification we will arrange the bad conductor in the following ways, we will take the cardboard having the dimensions of 5cm. The each side will have length of 5 cm (i.e a 5 cm square). Now the innerside is carved out having a dimension of 4 cm, similarly in other cardboards we will carve out the 3 cm, 2 cm, 1.5cm. After that we will place a very bad conductor between them which will completely cover the module.

First we will start the experiment with the cardboard in which the inner 4cm is carved out, this cardboard is placed between the Peltier module and the flame. The corresponding measurements are taken.

Now the experiments is repeated for all other bad insulators in which the inner 3cm, 2cm 1.5 cm is carved out and lastly the bad conductor itself in which no other dimensions are carved out.

The corresponding time – voltage graph is drawn for the insulators having different observations.

OBSERVATIONS (4 CM CARVED OUT)

TIME (s)	OBSERVED VOLTAGE (mV)
1	32
2	34
3	36
4	40
5	44
6	47

OBSERVATIONS (3 CM CARVED OUT)

TIME (s)	OBSERVED VOLTAGE (mV)
1	25
2	26
3	28
4	29
5	30
6	32

OBSERVATIONS (2 CM CARVED OUT)

TIME (s)	OBSERVED VOLTAGE (mV)
1	17
2	18
3	19
4	22
5	23

OBSERVATION (1.5 CM CARVED OUT)

TIME (s)	OBSERVED VOLTAGE (mV)
1	8
2	10
3	13
4	15

Now the complete insulator is placed between the Peltier module, even though little bit of heat is even transferred to module, the corresponding is as follows:

TIME (s)	OBSERVED VOLTAGE (mV)
1	2
2	3
3	4
4	6

After this value the output voltage was constant there wasn't a drastic change in the output voltage, hence we can assume that cardboard piece acts as a bad conductor of heat.

The corresponding time- voltage graph is drawn for each of the dimensions of the bad conductors.

SUNLIGHT – VOLTAGE STUDY

AIM

To study the Sunlight Voltage study using Peltier module and generated output is measured.

APPARATUES REQUIRED

Peltier Module, Convex Lens, Lens Holder, Multimeter

PROCEDURE

The Peltier module gives output voltage when heat is supplied to it, now we will study how the output voltage is related to the distribution of sunlight to a convex lens are of the Peltier module.

For that we use a Convex Lens to focus the incoming sunlight to the central most region of the Peltier Module.The experiment is conducted by placing the Convex Lens in a lens holder and focus the sunlight at various instances, starting from 10Am in the morning to 4PM in the evening.

The experiment is also conducted by placing the experimental set up in 3 different places where the intensity of sunlight reaching varies.

Starting from placing the experimental set up in the top floor of a multi-storied building which is approximately at a height of 25 meters from ground. Using the lens the sunlight is allowed to focus on to the Peltier module. Using a Multimeter connected to the both terminals of Pelteir module the Voltage output corresponding to various timings is measured.

Similarly, the experiment is repeated at an open ground too and corresponding output is measured.

Here we are plotting a graph between the observed output and timings.

OBSERVATIONS (25 m)

TIME	VOLTAGE**(Mv)**
10.00 AM	**4.5**
11.00 AM	**12.5**
12.00PM	**21.3**
1.00 PM	**35.3**
2.00 PM	**50.1**
3.00 PM	**45.2**
4.00 PM	**32.1**

OBSERVATIONS (Ground)

TIME	VOLTAGE**(Mv)**
10.00 AM	**3.1**
11.00 AM	**10.8**
AM	**20.01**
1.00 PM	**33.6**
2.00 PM	**47.9**
3.00 PM	**40.8**
4.00 PM	**29.3**

CHAPTER: 4

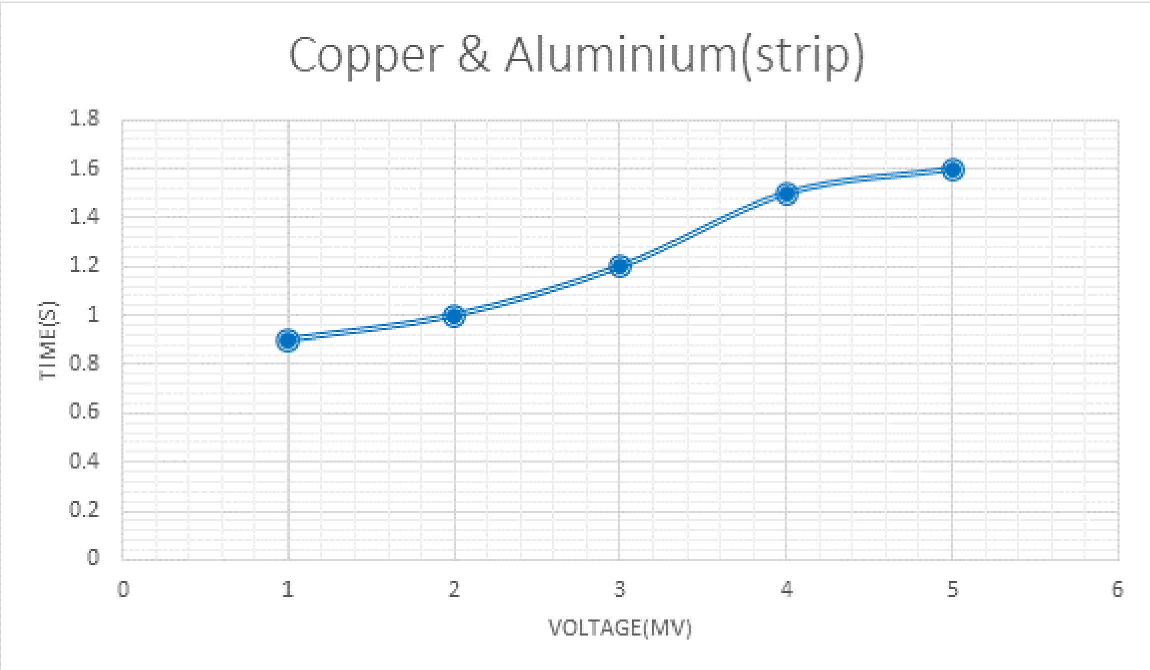
RESULTS AND DISCUSIONS

So far we have done experiments to verify the Seebeck effect and the Peltier module, from the experiments we have inferred some of the readings and some characteristics of the module has been understood.

Now we are going to infer the corresponding observations;

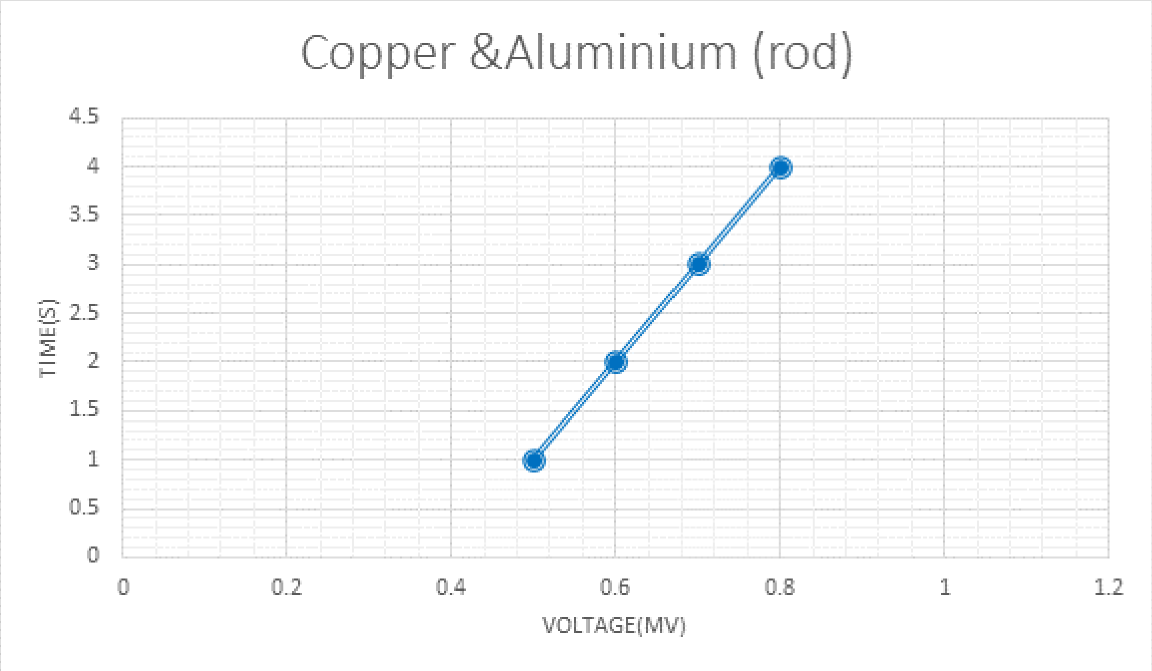
Our first experiment was to verify the Seebeck effect with different conducting materials. These were the observations noted.

For copper and aluminium (strip) the corresponding output graph is like this:



From the above graph we can see that as time increases, the output voltage(Mv) is also seen to be increased, from this we can understand that the Seebeck effect is true.

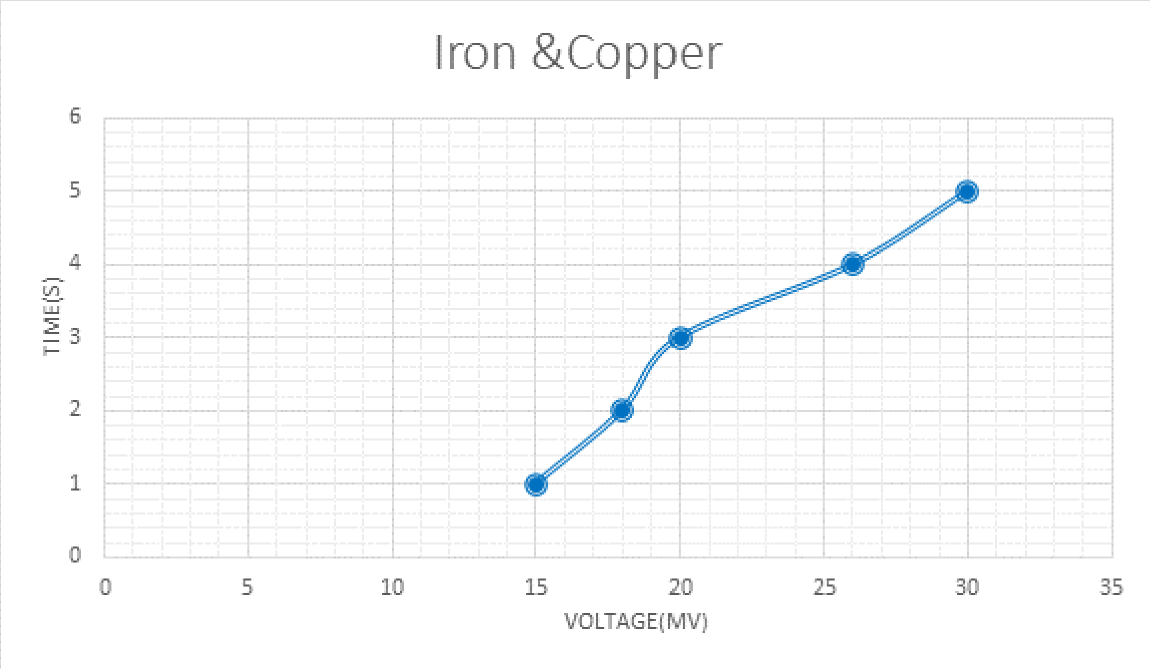
Now we are using copper and aluminum (rod) the corresponding graph is like this :



From the above two graphs in which aluminum in strip is used and in other experiment aluminum in rod is used from this we can see that the output voltage produced is more in the experiment in which the aluminum strip is used. When we are using the aluminum strip the maximum output voltage we get is more than the experiment involved in aluminum rod. This due to aluminum strip is broader and thin when compared to the rigid aluminum rod hence the thermal conductivity of strip is more than the thermal conductivity of the rod.

Moreover than that the strip shaped metals will have more loose electrons because they are in disformed shape due to this there is a surface defect in the strip like metals as a result the number of free electrons are more which will contribute towards the thermal conductivity of the metals. But in the case of rod shaped aluminum it is in perfect shape and it consists of rigid body shape in which the electrons are tightly bounded as a result the number of free electrons which take part in the reaction is less when we are comparing the number of free electrons in thin shaped aluminum.

Now we are using copper and iron in our experiment for the verification of Seebeck effect, the corresponding output graph is like this :



From the graph we can see that when we are using the iron and copper as the two distinct materials which are tied together and a junction is formed between them when we supply heat to this junction we can see that the output voltage is more, when we are comparing the other two experiments because the number of free electrons in copper and iron is more when we are comparing the number of free electrons in other metals , hence this is the reason for the high thermal conductivity shown by these metals as a result due to the flow of electrons we will get the output voltage. As time increases the rate of output voltage increases, but the supply of heat can't be given continuously, because as temperature increases slowly diffusion process takes place in the case of metals. Diffusion is the flow of metals from one place to another due to the vacancies in them.

When we are analyzing the tip of our metal pieces after sometime, we can see that the part of iron is in copper and vice versa, aluminum doesn't undergo diffusion easily this the reason why aluminum is used in IC's (integrated chips) because in high temperature's only aluminum undergoes diffusion.

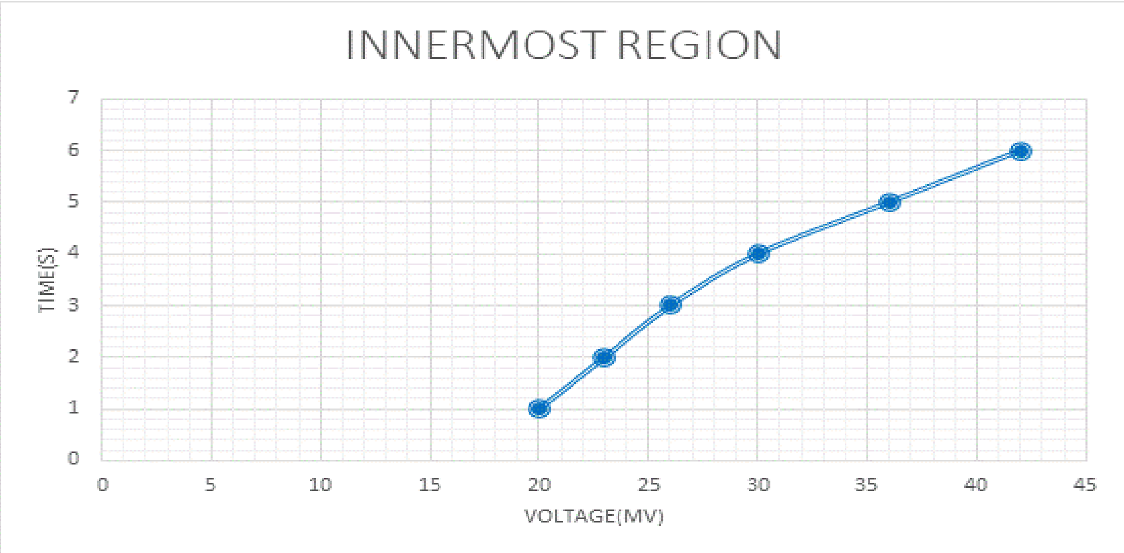
From the above experiments in which we verified the Seebeck effect and we found out the Seebeck voltage.

We already know that the Peltier module is a device which works on the Principle of Seebeck effect we already done a series of experiments to verify the working of the Peltier module from this the following inferences have been made out.

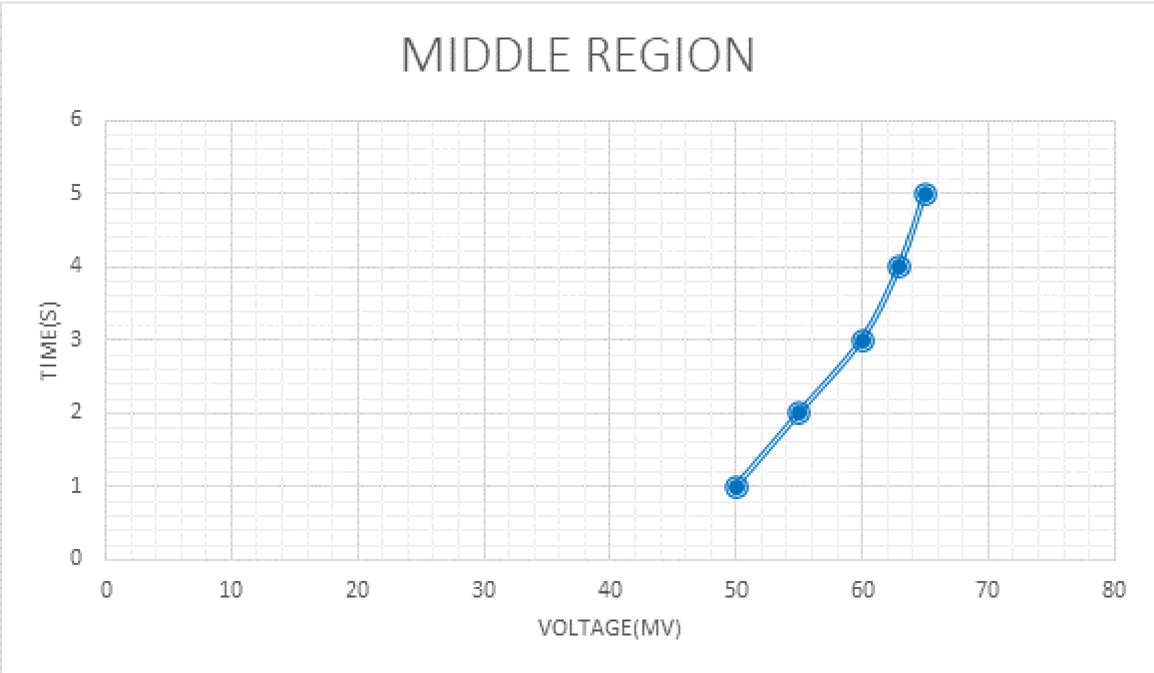
In our first experiment which was the **flame – displacement study** we arranged the Peltier module in three different zones of the temperature the following were the three different zones ;

1. Innermost region
2. Middle region
3. Outermost region

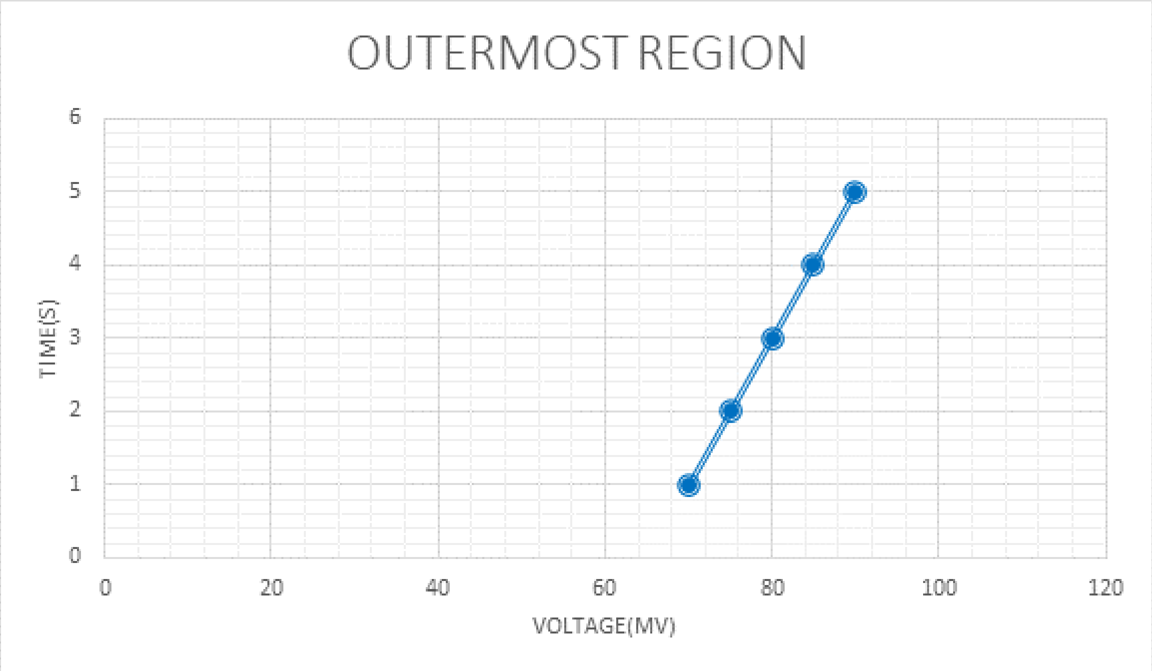
In the innermost region the corresponding graph we obtained is :



In the middle region the graph obtained is like this:



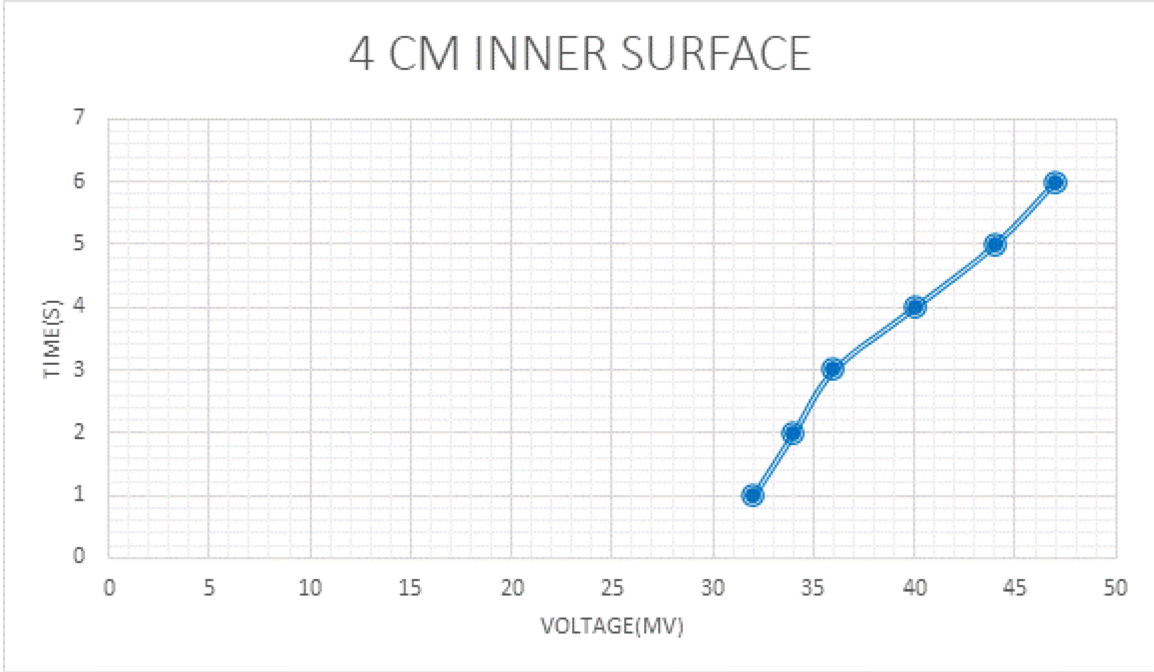
In the outermost region the graph obtained is like this :



From the above three graphs we obtained we can see that the output voltage we get from the outermost region is more when we are comparing the output voltages from the inner most region and the middle region. This is because in the outermost region the maximum combustion is taking place. The voltage that we get from the innermost region is very low when we are comparing the voltage from the middle region. The graph of the outermost region we are obtaining a straight line which shows that the voltage increase as time increases that is voltage is directly proportional to the output voltage.

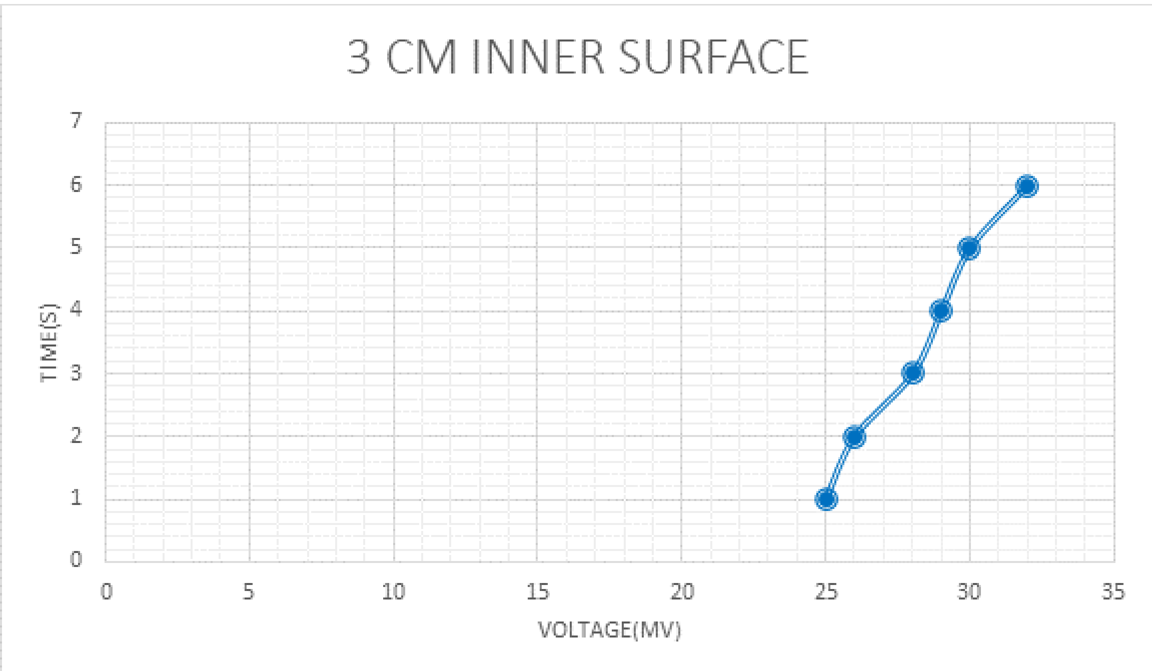
The second experiment we used **surface area and the corresponding voltage output :**

For the 4 cm inner surface of the Peltier we get the corresponding graph like this :

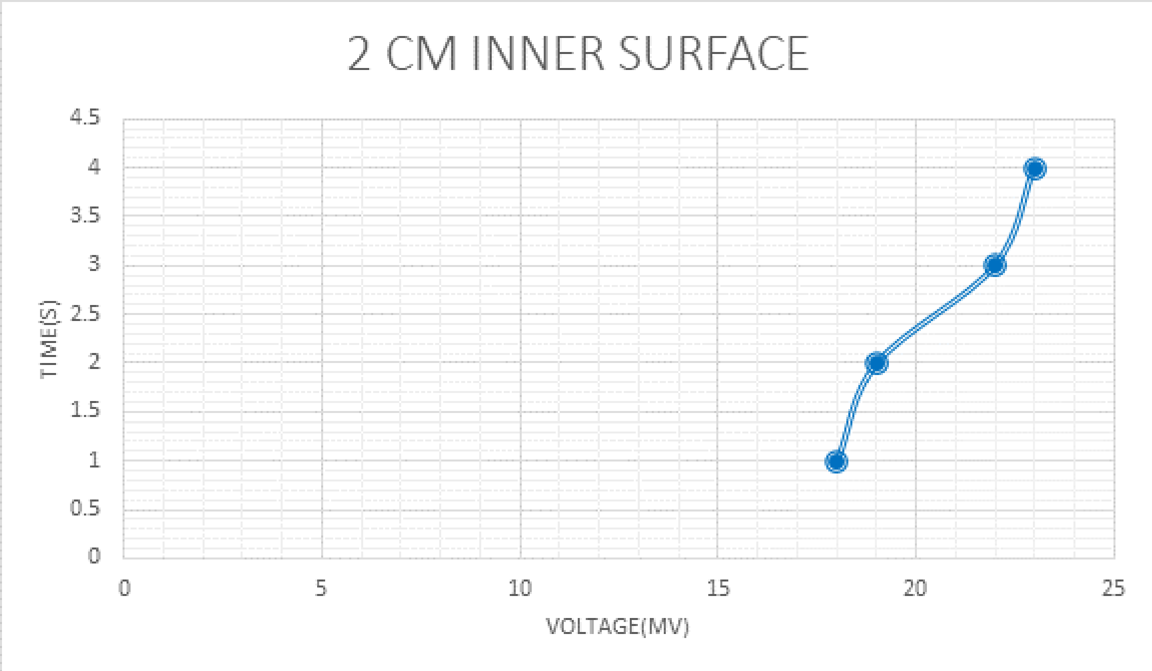


From this reading we can see that as time increase the corresponding output voltage also increases there is large area of module is visible to the heat as a result the corresponding output voltage is also high.

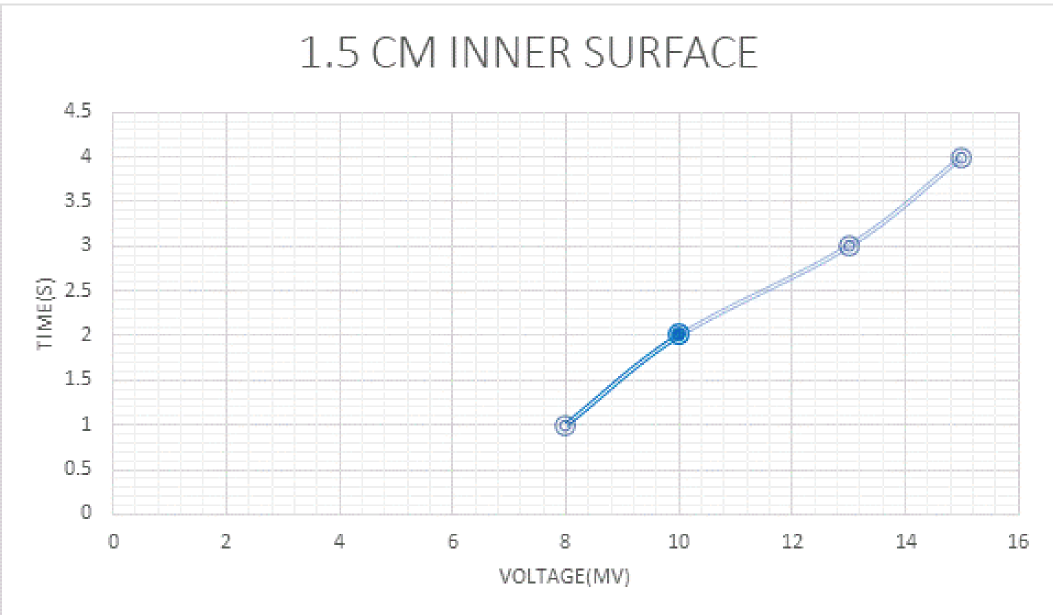
For the 3 cm inner surface the corresponding output voltage graph is like this:



For the 2 cm inner surface we will get the corresponding output voltage graph like this:



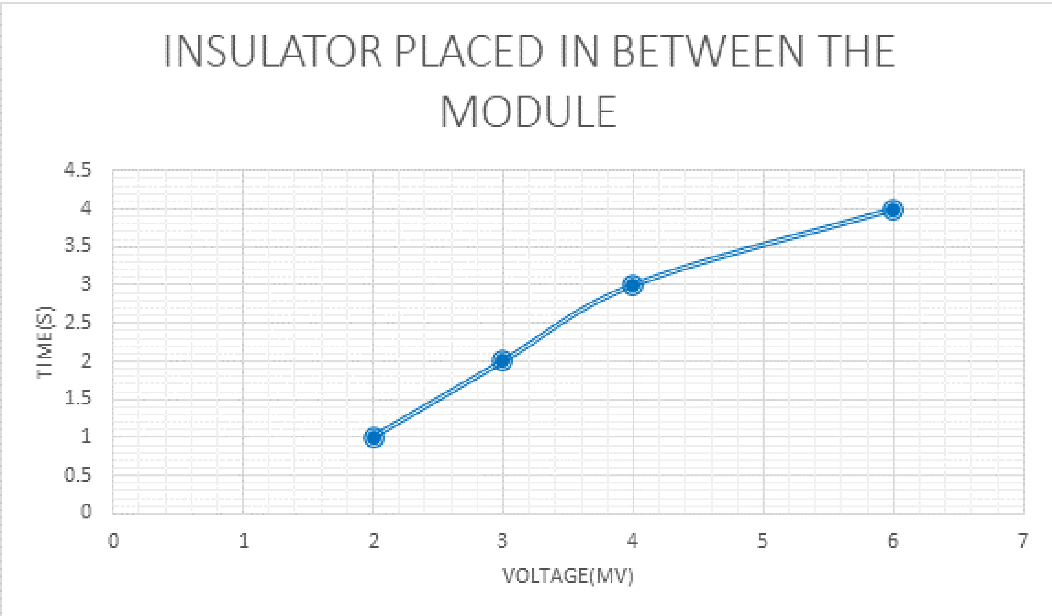
For the 1.5 cm inner surface we will get the corresponding voltage output graph like this :



From the above four graphs we can see that as the inner surface are varies the corresponding voltage output also varies, this is because when more heat is supplied to the larger surface area in the Peltier module more output voltage is produced. This is because the contact of heat is more, in larger surface area as a result more voltage is produced.

From this we can understand that as the temperature or heat increases the corresponding voltage is produced because the electrons move in one element and positive holes move in the other element. As a result when heat is suppled to larger surfaces more movement of electrons takes place. When we are using the small inner surface area for the module the corresponding voltage output is also low when compared to the larger inner surface area readings.

When we are using a bad conductor which have a equal surface area with the module and when we apply the heat the corresponding output graph is like this:

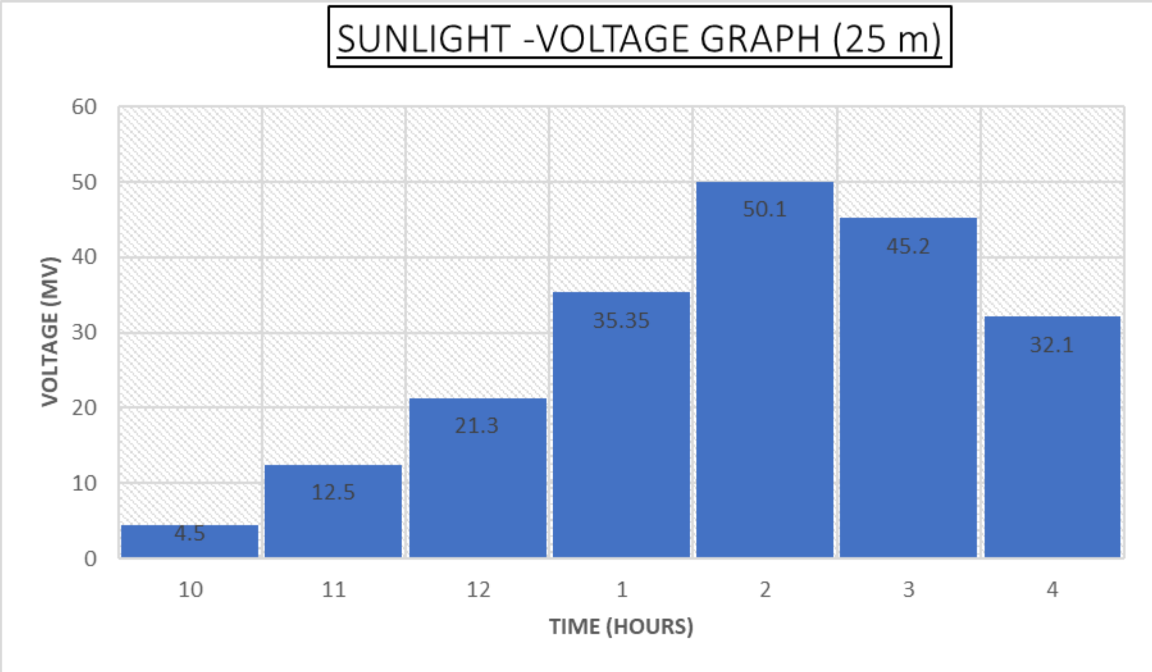


Here the maximum output voltage that we have received is very low from this we can see that there is only minimum amount of heat is reaching the bad conductor hence the corresponding output voltage is also low.

From the above experiments we can say that the output voltage produced is directly proportional to the surface area. The more the surface area the more the output voltage produced.

When we are using a bad conductor between the module and heat the corresponding output voltage is negligibly low hence we can say that if the amount of heat reaching the module is low, then the corresponding output voltage produced by Peltier module is also low. So heat is one of the factor which determines the voltage produced in the Peltier module.

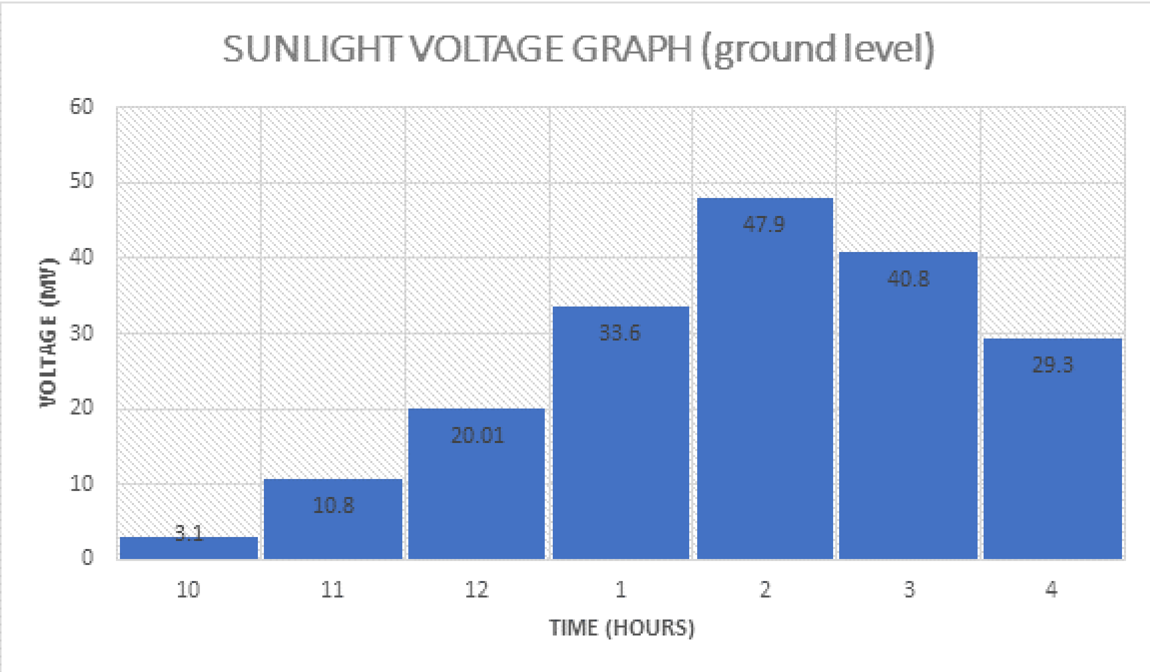
The third experiment was **sunlight – voltage study** the readings were taken from two different situations first was from an altitude of above 25m , (approx.) the second readings were taken in the ground level. The corresponding voltage we



obtained in the altitude of 25 m is as follows:

From this we can say that during the peak hours of temperature we received maximum output voltage at 10.00 am we received only the voltage of about 4.5 mv but as time increased the corresponding output voltage was also increased, due to the increase of intensity of sunlight which is a form of heat, in the atmosphere, during the peak noon time there was an tremendous increase in the voltage level around 2.00 pm the voltage was around 50.1mv because of the hotness of the atmosphere and the corresponding temperature gradient was formed in module. But as time goes by the corresponding output voltage also decreases due to the atmospheric temperature decreases. We can see that the around 4.00 pm the output voltage is less, when we are comparing the output voltages of the 2.00 pm ,3.00 pm, there is a notable difference in their output voltages.

When we placed the module in the ground level the graph obtained was like this:



From this we can see that there is also a increase in the output voltage as time goes by during 10.00 am the output voltage is 3.1 mv but as time increases we get the output voltage 47.9 mv in 2.00 pm but it also gets decreases as time goes and around 4.00 pm the output voltage is 29.3mv, this also shows that there is a increase and decrease of voltage when we are supplying atmospheric sunlight to the module.

From the above two series of experiments one in an altitude and in the ground level and when we compare it we can see that the output voltage corresponding to higher altitude region is more than the output voltage produced in the ground level. This is because when we are in an elevated position, we are close to the sun when compared to the ground level there is a small difference between these two levels, this difference can be neglected when we are comparing it with the distance of us to the sun. Here the rays coming from sun undergoes refraction with the atmospheric particles, Relative degree of refraction at higher altitude is less than the refraction at lower altitude due to this the intensity of sunlight is different in two different heights. As a result we will get more intense sunlight at 25 m altitude and less sunlight in the ground level as a result the corresponding temperature gradient formed in this regions are different.

From the above observations and their corresponding results we van finalize that the Peltier module can be used as an alternating source for the production of the voltage. A single Peltier module can only produce low range voltage as output but when we are using numerous or large number of Peltier components we can get more voltage as the output.

These voltages that we get from this Peltier module can be used to charge the electronical devices, can be used as a source voltage for the working of many devices. We also mentioned that the Peltier module is thermoelectric cooler which can cbe used in DC voltages this will give out heat in one side and the other side will become cooler even below the room temperature.

CHAPTER: 5

CONCLUSION

From the following series of experiments we have dine and the corresponding observations and results we can conclude that Peltier module can be used as a alternative energy source. This output voltage we get from a Peltier module is comparatively low but when we supply this output voltage to a voltage multiplier or corresponding voltage multiplying devices (Bug Booster – Step up) we can get multiplied output voltage, then this voltage can be utilized to supply the working voltages for other electronic devices.

ADVANTAGES OF THE PELTIER MODULE

1. The Peltier module is used by excavators, archaeologists and so many other people for their use. When they go deep into jungle and other places where there is no electricity they will the Peltier module to produce the voltage and this voltage can be used to enable the working of the electronic devices.
2. The Peltier module is a thermoelectric cooler in which , by applying a proper DC voltage we will give heat in one side and the other side will be cooler. As a result the Peltier module can be used to make a mini refrigerator which can cool elements below than the room temperature.
3. The peltier module can be used to give heat on one side and cool atmosphere on the opposite side, so roof made up of Peltier components can be used as a roofing materials on the top of the houses, as a result we can control the atmospheric conditions in the house. The Peltier components roof can be made and be installed in cold weather countries where the temperature is always below than zero degree celsuis, as a result the roof side will be cooler and the corresponding inner side will be hotter, when we apply a proper DC voltage to it.
4. Peltier module can be used for temperature control where proper control of the temperature is required. Such as in laboratory equipments, medical devices, and temperature- controlled enclosures for electronics.
5. Peltier modules can be used in car seat coolers/ heaters or in electronic cabinets to prevent overheating.
6. Thermal Cycling : Peltier modules are used in thermal cycles for polymerase chain reaction (PCR) machines, which require precise and rapid heating and cooling cycles for DNA amplification.
7. Waste heat recovery : Peltier module can be used to recover waste heat from various sources, such as industrial processes or vehicle exhaust, and convert it into usable electrical energy. Using this output voltage we will be able to work electronic devices.

FUTURE OF PELTIER MODULES

1. **Increased efficiency** :Researchers are continuously working on improving the efficiency of Peltier modules to make them more energy- efficient, allowing for better cooling/heating performance while minimising power consumption.
2. **Materials innovation** : Reaserch into novel thermoelectric materials with improved efficiency, stability and constant effectiveness, is on going. This includes exploring new compounds, nano-structured materials, and composites to enhance the performance of Peltier modules
3. So much researches are still ongoing on this field so much is planned to carry on Peltier module has it's vast opportunities which can be applied in the field of space technologies, and related fileds. It has more opportunities in the field of medical field and atomic researches.

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